INCH POUND

MIL-C-3098/109C
5 May 1993
SUPERSEDING
MIL-C-3098/109B
23 June 1975

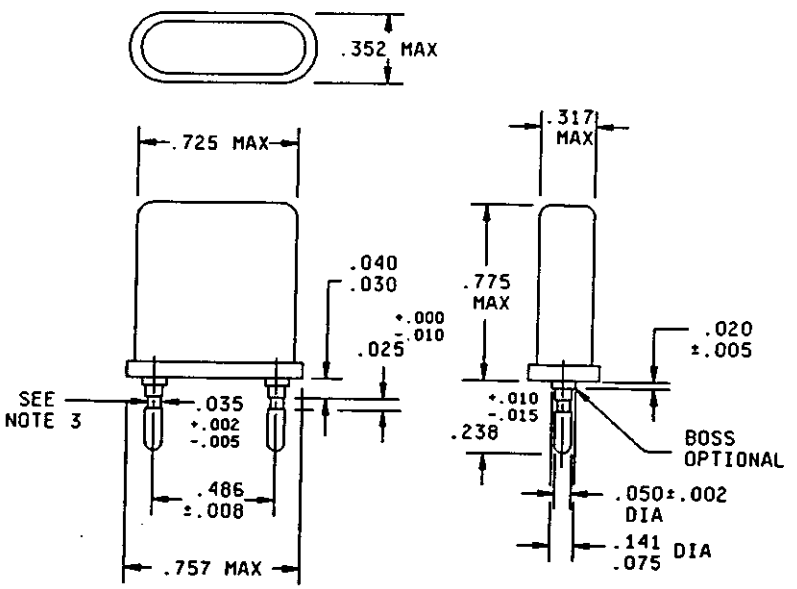
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR130/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 0.8 to 20 MHz; fundamental; noncontrolled; antiresonance.



Inches	mm	Inches	mm
.002	.05	.050	1.27
.005	.13	.075	1.91
.008	.20	.141	3.58
.010	.25	.238	6.05
.015	.38	.317	8.05
.020	.51	.352	8.94
.025	.64	.486	12.34
.030	.76	.725	18.42
.035	.89	.757	19.23
.040	1.02	.775	19.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be in accordance with MIL-C-3098.
4. The pin undercut may be omitted.

FIGURE 1. Crystal unit - CR130/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Holder: HC-6/U (MIL-H-10056/2). 1/

Dimensions, marking, and configuration: See figure 1.

Frequency range: 0.8 to 20 MHz, inclusive.

Frequency tolerance (operating temperature range):

Primary: ±20 ppm.

Secondary: ±30 ppm.

Equivalent resistance: See table I.

TABLE I. Equivalent resistance.

Frequency range, inclusive			Maximum resistance
MHz			Ohms
0.8	to	0.85	620
0.85+	to	0.9	600
0.9+	to	1	570
1+	to	1.12	540
1.12+	to	1.25	490
1.25+	to	1.37	450
1.37+	to	1.5	410
1.5+	to	1.62	370
1.62+	to	1.75	330
1.75+	to	1.87	300
1.87+	to	2.0	290
2.0+	to	2.12	270
2.12+	to	2.25	240
2.25+	to	2.6	190
2.6+	to	3	150
3+	to	3.4	110
3.4+	to	3.75	90
3.75+	to	4	75
4+	to	5	60
5+	to	7	35
7+	to	10	24
10+	to	15	22
15+	to	20	20

Mode of oscillation: Fundamental.

Temperature ranges:

Primary: -40°C to +90°C, inclusive.

Secondary: -55°C to -40°C and +90°C to +105°C, inclusive.

Drive level: 0.8 to 10.0 MHz = 10.0 ±2.0 mW.
10.0+ to 20.0 MHz = 5.0 ±1.0 mW.

Capacitance, shunt: 7.0 pF maximum.

Antiresonance load capacitance: 30 ±0.5 pF. 2/

1/ The socket designed to mate with crystal holder HC-6/U is designated in MIL-S-12883/4 (Sockets, Crystal, Top and Bottom Mounting, 2 Contacts). For Navy use, other sockets listed in MIL-STD-242 may be used.
2/ This crystal unit is similar to Crystal Type CR-85/U, except for operation at antiresonance and maximum equivalent resistance.

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Shock (specified pulse):	0.8 to 2.0 MHz	2.0+ to 20.0 MHz
Frequency change permitted:	±10 ppm	±5 ppm
Equivalent resistance change permitted:	±15 percent	±10 percent
Vibration: Method 201 of MIL-STD-202, test condition A.	0.8 to 2.0 MHz	2.0+ to 20.0 MHz
Frequency change permitted:	±10 ppm	±5 ppm
Equivalent resistance change permitted:	±15 percent	±10 percent
Thermal shock:	0.8 to 2.0 MHz	2.0+ to 20.0 MHz
Frequency change permitted:	±10 ppm	±5 ppm
Equivalent resistance change permitted:	±15 percent	±10 percent
Aging:		
Frequency change permitted: ±5 ppm.		

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:

Army - ER
Navy - EC
Air Force - 85

Review activities:

Army - AR, MI, SM

User activities:

Navy - AS, MC, SH
Air Force - 17, 19, 84

Preparing activity:

Army - ER

Agent:

DLA - ES

(Project 5955-0678-06)

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